

Esercizio 1. Risolvere l'equazione alle differenze

$$\Delta^2 (\Delta - 1) y(t) = 3 \delta_0(t)$$

imponendo il passaggio per le condizioni iniziali ed evidenziando risposta libera e forzata

La soluzione generale dell'omogenea è

$$y_{g,o}(t) = c_1 \delta_0(t) + c_2 \delta_0(t-1) + c_3$$

La soluzione particolare è risonante

La risonanza fa nascere un termine aggiuntivo il cui “grado” è

$$\text{grado ingresso} + \text{grado soluzione omogenea} + 1$$

$$y_p(t) = \alpha_1 \cancel{\delta_0}(t) + \alpha_2 \delta_0(\cancel{t}-1) + \alpha_3 \delta_0(t-2)$$

Per calcolare α_3 , sostituisco y_p nell'equazione

$$\begin{aligned} \Delta^3 y_p(t) - \Delta^2 y_p(t) &= 3 \delta_0(t) \\ \alpha_3 \delta_0(\cancel{t}+1) - \alpha_3 \delta_0(t) &= 3 \delta_0(t) \\ \alpha_3 &= -3 \end{aligned}$$

$$y_p(t) = -3 \delta_0(t-2)$$

La soluzione generale della non omogenea è

$$\begin{aligned} y_{g,no}(t) &= y_{g,o}(t) + y_p(t) \\ &= c_1 \delta_0(t) + c_2 \delta_0(t-1) + c_3 - 3 \delta_0(t-2) \end{aligned}$$

Impongo il passaggio per le condizioni iniziali

$$\begin{aligned} y(0) &= y_0 \\ c_1 + c_3 &= y_0 \\ y(1) &= y_1 \\ c_2 + c_3 &= y_1 \\ y(2) &= y_2 \\ c_3 - 3 &= y_2 \end{aligned}$$

$$\begin{aligned} c_3 &= y_2 + 3 \\ c_2 &= y_1 - c_3 \\ &= y_1 - y_2 - 3 \\ c_1 &= y_0 - c_3 \\ &= y_0 - y_2 - 3 \end{aligned}$$

$$\begin{aligned} y(t) &= \underbrace{(y_0 - y_2 - 3) \delta_0(t) + (y_1 - y_2 - 3) \delta_0(t-1) + (y_2 + 3)}_{\text{soluzione pseudo-permanente}} \underbrace{- 3 \delta_0(t-2)}_{\text{componente pseudo-transitoria}} \\ &= \underbrace{(y_0 - y_2) \delta_0(t) + (y_1 - y_2) \delta_0(t-1) + (y_2)}_{\text{soluzione libera}} \underbrace{- 3 \delta_0(t-2) - 3 \delta_0(t) - 3 \delta_0(t-1) + 3}_{\text{soluzione forzata}} \end{aligned}$$

Esercizio 2. Risolvere l'equazione alle differenze

$$(\Delta^2 - \Delta + 1) y(t) = 2(-1)^t$$

imponendo il passaggio per le condizioni iniziali ed evidenziando risposta libera e forzata

Vado a determinare le radici del polinomio caratteristico poiché non è fattorizzato

$$\begin{aligned}\Delta_{1,2} &= \frac{1 \pm \sqrt{1-4}}{2} \\ &= \frac{1 \pm i\sqrt{3}}{2} \\ \rho &= \sqrt{\left(\frac{1}{2}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2} \\ &= \sqrt{\frac{1}{4} + \frac{3}{4}} = 1 \\ \omega &= \arctan\left(\frac{\frac{\sqrt{3}}{2}}{\frac{1}{2}}\right) \\ &= \arctan(\sqrt{3}) \\ &= \frac{\pi}{3}\end{aligned}$$

La soluzione generale dell'omogenea è

$$y_{g,o}(t) = c_1 \cos\left(\frac{\pi}{3}t\right) + c_2 \sin\left(\frac{\pi}{3}t\right)$$

Non ho risonanza

$$y_p(t) = \alpha (-1)^t$$

$$\begin{aligned}\text{se } u(t) &= (-1)^t \\ \bar{\alpha} &= \frac{1}{p(-1)} \\ &= \frac{1}{(-1)^2 - (-1) + 1} \\ &= \frac{1}{3}\end{aligned}$$

$$\begin{aligned}\alpha &= 2\bar{\alpha} \\ &= \frac{2}{3}\end{aligned}$$

$$y_{g,no}(t) = c_1 \cos\left(\frac{\pi}{3}t\right) + c_2 \sin\left(\frac{\pi}{3}t\right) + \frac{2}{3}(-1)^t$$

$$\begin{aligned}y(0) &= y_0 \\ c_1 + \frac{2}{3} &= y_0 \\ y(1) &= y_1 \\ c_1 \frac{1}{2} + c_2 \frac{\sqrt{3}}{2} - \frac{2}{3} &= y_1\end{aligned}$$

$$\begin{aligned}c_1 &= y_0 - \frac{2}{3} \\ c_2 &= \frac{2}{\sqrt{3}}\left(y_1 - \frac{1}{2}c_1 + \frac{2}{3}\right) \\ &= \frac{2}{\sqrt{3}}\left(y_1 - \frac{1}{2}\left(y_0 - \frac{2}{3}\right) + \frac{2}{3}\right)\end{aligned}$$

Esercizio 3. Si calcoli la soluzione del seguente sistema, evidenziando risposta libera, risposta forzata, risposta pseudo-permanente e risposta pseudo-transitoria

$$\dot{x}(t) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t$$

Calcolo la soluzione generale dell'omogenea

$$\begin{aligned} x_{g,o}(t) &= e^{At} c \\ e^{At} &= \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix} \\ x_{g,o}(t) &= \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \\ &= \begin{pmatrix} c_1 e^t \\ c_2 e^{-t} \end{pmatrix} \end{aligned}$$

Calcolo la soluzione particolare

$$\begin{aligned} x_p(t) &= e^t \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \\ \dot{x}_p(t) &= A x(t) + B u(t) \\ e^t \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} &= e^t \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t \\ \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \right) \cdot \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \end{aligned}$$

Il sistema non ha soluzione $\Rightarrow$ ho sbagliato la soluzione particolare

$$\begin{aligned} x_p(t) &= e^t \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + t e^t \begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} \\ e^t \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + (t e^t + e^t) \begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} &= e^t \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + t e^t \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} + \begin{pmatrix} e^t \\ 0 \end{pmatrix} \\ \begin{pmatrix} \alpha_3 t e^t + \alpha_3 e^t + \alpha_1 e^t \\ \alpha_4 t e^t + \alpha_4 e^t + \alpha_2 e^t \end{pmatrix} &= \begin{pmatrix} \alpha_3 t e^t + \alpha_1 e^t + e^t \\ -(\alpha_4 t e^t) - \alpha_2 e^t \end{pmatrix} \end{aligned}$$

$$\begin{aligned} \alpha_3 &= \alpha_3 \\ \alpha_3 + \alpha_1 &= \alpha_1 + 1 \\ \alpha_4 &= -\alpha_4 \\ \alpha_4 + \alpha_2 &= -\alpha_2 \end{aligned}$$

$$\begin{aligned} \alpha_3 &= 1 \\ \alpha_4 &= 0 \\ \alpha_2 &= 0 \\ \alpha_1 &\text{ è arbitrario (=0)} \end{aligned}$$

$$x_p(t) = t e^t \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$x_{g, \text{no}}(t) = \underbrace{\begin{pmatrix} c_1 e^t \\ c_2 e^{-t} \end{pmatrix}}_{\text{componente pseudo-transitoria}} + \underbrace{t e^t \begin{pmatrix} 1 \\ 0 \end{pmatrix}}_{\text{soluzione pseudo-permanente}}$$

$$x(0) = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = x_0$$

$$x(t) = \underbrace{\begin{pmatrix} x_{0,1} e^t \\ x_{0,2} e^{-t} \end{pmatrix}}_{\text{risposta libera}} + \underbrace{\begin{pmatrix} t e^t \\ 0 \end{pmatrix}}_{\text{risposta forzata}}$$

Variante

$$\dot{x}(t) = A x(t) + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^t t$$

$$x_p(t) = e^t t \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + e^t \begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix}$$

$$\text{se } A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$x_{g,o} = \begin{pmatrix} e^t c_1 \\ e^{-t} c_2 \end{pmatrix}$$

$$x_p(t) = e^t t \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + e^t \begin{pmatrix} \alpha_3 \\ \alpha_4 \end{pmatrix} + e^t t^2 \begin{pmatrix} \alpha_5 \\ \alpha_6 \end{pmatrix}$$

Esercizio 4. Si calcoli la soluzione del seguente sistema, evidenziando risposta libera, risposta forzata, risposta pseudo-permanente e risposta pseudo-transitoria

$$x(t+1) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ 0 \end{pmatrix} (-2)^t$$

Calcolo

$$\begin{aligned} p(s) &= \det(sI - A) \\ &= \det\left(s \text{ ident}(2) - \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}\right) \\ &= \det\begin{pmatrix} s-1 & -1 \\ 0 & s+1 \end{pmatrix} \\ &= (s-1)(s+1) \\ \sigma(A) &= \{1, -1\} \end{aligned}$$

$$\begin{aligned} \lambda_1 &= 1 \\ \text{Ker}(\lambda_1 I - A) &= \text{Ker}\begin{pmatrix} 1-1 & -1 \\ 0 & 1+1 \end{pmatrix} \\ &= \text{Ker}\begin{pmatrix} 0 & -1 \\ 0 & 2 \end{pmatrix} \\ \begin{pmatrix} 0 & -1 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} &= 0 \end{aligned}$$

$$\begin{aligned}
\begin{pmatrix} -b \\ 2b \end{pmatrix} &= 0 \\
&= \text{Span}\left(\begin{pmatrix} 1 \\ 0 \end{pmatrix}\right) \\
v_1 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
\lambda_2 &= -1 \\
\text{Ker}(\lambda_2 I - A) &= \text{Ker}\begin{pmatrix} -1-1 & -1 \\ 0 & -1+1 \end{pmatrix} \\
&= \text{Ker}\begin{pmatrix} -2 & -1 \\ 0 & 0 \end{pmatrix} \\
\begin{pmatrix} -2 & -1 \\ 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} a \\ b \end{pmatrix} &= 0 \\
\begin{pmatrix} -b-2a \\ 0 \end{pmatrix} &= 0 \\
b &= -2a \\
\text{Ker}(\lambda_2 I - A) &= \text{Span}\left(\begin{pmatrix} 1 \\ -2 \end{pmatrix}\right) \\
v_2 &= \begin{pmatrix} 1 \\ -2 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
V &= \begin{pmatrix} 1 & 1 \\ 0 & -2 \end{pmatrix} \\
V^{-1} &= \frac{1}{-2} \begin{pmatrix} -2 & -1 \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & -(\frac{1}{2}) \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
E_1 &= \begin{pmatrix} 1 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 & \frac{1}{2} \end{pmatrix} \\
&= \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{pmatrix} \\
E_2 &= \begin{pmatrix} 1 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 0 & -(\frac{1}{2}) \end{pmatrix} \\
&= \begin{pmatrix} 0 & -(\frac{1}{2}) \\ 0 & 1 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
A &= 1 \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{pmatrix} + (-1) \begin{pmatrix} 0 & -(\frac{1}{2}) \\ 0 & 1 \end{pmatrix} \\
A^t &= 1^t \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{pmatrix} + (-1)^t \begin{pmatrix} 0 & -(\frac{1}{2}) \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & \frac{1}{2} - \frac{(-1)^t}{2} \\ 0 & (-1)^t \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
\frac{1}{p(s)} &= \frac{1}{(s+1)(s-1)} \\
&= \frac{A}{s+1} + \frac{B}{s-1} \\
A &= \text{subst}\left(s=-1, \frac{1}{s-1}\right) \\
&= -\frac{1}{2} \\
B &= \text{subst}\left(s=1, \frac{1}{s+1}\right) \\
&= \frac{1}{2} \\
\frac{1}{p(s)} &= \frac{-\frac{1}{2}}{s+1} + \frac{\frac{1}{2}}{s-1}
\end{aligned}$$

$$\begin{aligned}
f_1(s) &= \frac{-\frac{1}{2}}{s \not= 1} \frac{(s \not= 1)(s-1)}{(s+1)(s-1)} \\
f_2(s) &= \frac{\frac{1}{2}}{s \not= 1} \frac{(s+1)(s \not= 1)}{(s+1)(s-1)}
\end{aligned}$$

$$\begin{aligned}
e_1(s) &= -\frac{1}{2}(s-1) \quad \lambda_1 = -1 \\
e_2(s) &= \frac{1}{2}(s+1) \quad \lambda_2 = +1
\end{aligned}$$

$$\begin{aligned}
E_1 &= -\frac{1}{2}(A-I) \\
&= -\frac{1}{2}\left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} - \text{ident}(2)\right) \\
&= \begin{pmatrix} 0 & -(\frac{1}{2}) \\ 0 & 1 \end{pmatrix} \\
E_2 &= \frac{1}{2}(A+I) \\
&= \frac{1}{2}\left(\begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} + \text{ident}(2)\right) \\
&= \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{pmatrix} \\
A^t &= 1^t \begin{pmatrix} 1 & \frac{1}{2} \\ 0 & 0 \end{pmatrix} + (-1)^t \begin{pmatrix} 0 & -(\frac{1}{2}) \\ 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & \frac{1}{2} - \frac{(-1)^t}{2} \\ 0 & (-1)^t \end{pmatrix} \\
x_{g,o}(t) &= \begin{pmatrix} 1 & \frac{1}{2} - \frac{(-1)^t}{2} \\ 0 & (-1)^t \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
x_p(t) &= (-2)^t \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} \\
x_p(t+1) &= A x_p(t) + B u(t) \\
(-2)^{t+1} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} &= (-2)^t \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + (-2)^t \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
(-2) (-2)^t \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} &= (-2)^t \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + (-2)^t \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
-2 \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} &= \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix} \cdot \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \\
\begin{pmatrix} -(2\alpha_1) \\ -(2\alpha_2) \end{pmatrix} &= \begin{pmatrix} \alpha_2 + \alpha_1 + 1 \\ -\alpha_2 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
\alpha_2 &= 0 \\
-2\alpha_1 - \alpha_1 &= 1 \\
\alpha_1 &= -\frac{1}{3}
\end{aligned}$$

$$x_p(t) = (-2)^t \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}$$

$$x_{g, \text{no}}(t) = \begin{pmatrix} 1 & \frac{1}{2} - \frac{(-1)^t}{2} \\ 0 & (-1)^t \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + (-2)^t \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}$$

$$x(0) = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}$$

$$c = x_0 - \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}$$

$$= x_0 + \begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix}$$

$$\begin{aligned}
x(t) &= \underbrace{\begin{pmatrix} 1 & \frac{1}{2} - \frac{(-1)^t}{2} \\ 0 & (-1)^t \end{pmatrix} \cdot \left(x_0 + \begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix} \right)}_{\text{componente pseudo-transitoria}} + \underbrace{(-2)^t \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}}_{\text{soluzione pseudo-permanente}} \\
&= \underbrace{\begin{pmatrix} 1 & \frac{1}{2} - \frac{(-1)^t}{2} \\ 0 & (-1)^t \end{pmatrix} x_0}_{\text{risposta libera}} + \underbrace{\begin{pmatrix} 1 & \frac{1}{2} - \frac{(-1)^t}{2} \\ 0 & (-1)^t \end{pmatrix} \cdot \begin{pmatrix} \frac{1}{3} \\ 0 \end{pmatrix} + (-2)^t \begin{pmatrix} -\frac{1}{3} \\ 0 \end{pmatrix}}_{\text{risposta forzata}}
\end{aligned}$$

Esercizio 5. Si calcoli la soluzione del seguente sistema, evidenziando risposta libera, risposta forzata, risposta pseudo-permanente e risposta pseudo-transitoria

$$\dot{x}(t) = \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix} x(t) + \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cos(t)$$

Per calcolare la soluzione generale dell'omogenea devo calcolare e^{At}

$$\begin{aligned}
 p(s) &= \det(sI - A) \\
 &= \det\left(s \text{ident}(3) - \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix}\right) \\
 &= \det\begin{pmatrix} s-1 & -1 & 0 \\ 0 & s+1 & -2 \\ 0 & 0 & s-1 \end{pmatrix} \\
 &= (s-1)^2(s+1)
 \end{aligned}$$

$$\begin{aligned}
 \frac{1}{p(s)} &= \frac{1}{(s-1)^2(s+1)} \\
 &= \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{C}{s+1}
 \end{aligned}$$

$$\begin{aligned}
 B &= \text{subst}\left(s=1, \frac{1}{(s+1)}\right) \\
 &= \frac{1}{2}
 \end{aligned}$$

$$\begin{aligned}
 C &= \text{subs}\left(s=-1, \frac{1}{(s-1)^2}\right) \\
 &= \frac{1}{4}
 \end{aligned}$$

$$0 = A + C$$

$$\begin{aligned}
 A &= -C \\
 &= -\frac{1}{4}
 \end{aligned}$$

$$\text{partfrac}\left(\frac{1}{(s-1)^2(s+1)}, s\right) = \frac{1}{4(s+1)} - \frac{1}{4(s-1)} + \frac{1}{2(s-1)^2}$$

$$\begin{aligned}
 f_1(s) &= \frac{1}{4(s+1)} \frac{p(s)}{p(s)} \\
 &= \frac{1}{4(s \neq 1)} \frac{(s-1)^2 (s \neq 1)}{p(s)}
 \end{aligned}$$

$$\begin{aligned}
 f_2(s) &= \left(-\frac{1}{4(s-1)} + \frac{1}{2(s-1)^2}\right) \frac{p(s)}{p(s)} \\
 &= -\left(\frac{s-3}{4(s \neq 1)^2}\right) \frac{(s \neq 1)^2 (s+1)}{p(s)}
 \end{aligned}$$

$$\begin{aligned}
 e_1(s) &= \frac{1}{4}(s-1)^2 \\
 &= \frac{s^2}{4} - \frac{s}{2} + \frac{1}{4}
 \end{aligned}$$

$$\begin{aligned}
 e_2(s) &= -\frac{1}{4}(s-3)(s+1) \\
 &= -\left(\frac{s^2}{4}\right) + \frac{s}{2} + \frac{3}{4}
 \end{aligned}$$

$$E_1 = \frac{1}{4}A^2 - \frac{1}{2}A + \frac{1}{4}$$

$$\begin{aligned}
&= \frac{1}{4} \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix} + \frac{1}{4} \text{ident}(3) \\
&= \begin{pmatrix} 0 & -(\frac{1}{2}) & \frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \\
E_2 &= \text{ident}(3) - \begin{pmatrix} 0 & -(\frac{1}{2}) & \frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \\
&= \begin{pmatrix} 1 & \frac{1}{2} & -(\frac{1}{2}) \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
D &= \lambda_1 E_1 + \lambda_2 E_2 \\
&= -1 \begin{pmatrix} 0 & -(\frac{1}{2}) & \frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} + 1 \begin{pmatrix} 1 & \frac{1}{2} & -(\frac{1}{2}) \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
N &= A - D \\
&= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}
\end{aligned}$$

$$N^2 = 0$$

$$K = 2$$

$$\begin{aligned}
e^{At} &= e^{Dt} (I + tN) \\
&= \begin{pmatrix} e^t & \frac{e^t}{2} - \frac{e^{-t}}{2} & \frac{e^{-t}}{2} - \frac{e^t}{2} \\ 0 & e^{-t} & e^t - e^{-t} \\ 0 & 0 & e^t \end{pmatrix} \cdot \left(\text{ident}(3) + t \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right) \\
&= \begin{pmatrix} e^t & \frac{e^t}{2} - \frac{e^{-t}}{2} & t e^t - \frac{e^t}{2} + \frac{e^{-t}}{2} \\ 0 & e^{-t} & e^t - e^{-t} \\ 0 & 0 & e^t \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
e^{Dt} &= e^{-t} \begin{pmatrix} 0 & -(\frac{1}{2}) & \frac{1}{2} \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} + e^t \begin{pmatrix} 1 & \frac{1}{2} & -(\frac{1}{2}) \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} e^t & \frac{e^t}{2} - \frac{e^{-t}}{2} & \frac{e^{-t}}{2} - \frac{e^t}{2} \\ 0 & e^{-t} & e^t - e^{-t} \\ 0 & 0 & e^t \end{pmatrix}
\end{aligned}$$

$$x_p(t) = \cos(t) \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} + \sin(t) \begin{pmatrix} \alpha_4 \\ \alpha_5 \\ \alpha_6 \end{pmatrix}$$

$$\begin{aligned}
-\sin(t) \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} + \cos(t) \begin{pmatrix} \alpha_4 \\ \alpha_5 \\ \alpha_6 \end{pmatrix} &= \cos(t) \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} + \sin(t) \begin{pmatrix} 1 & 1 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \alpha_4 \\ \alpha_5 \\ \alpha_6 \end{pmatrix} + \\
&\quad \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cos(t) \\
\begin{pmatrix} \alpha_4 \cos(t) - \alpha_1 \sin(t) \\ \alpha_5 \cos(t) - \alpha_2 \sin(t) \\ \alpha_6 \cos(t) - \alpha_3 \sin(t) \end{pmatrix} &= \begin{pmatrix} (\alpha_5 + \alpha_4) \sin(t) + (\alpha_2 + \alpha_1) \cos(t) + \cos(t) \\ (2\alpha_6 - \alpha_5) \sin(t) + (2\alpha_3 - \alpha_2) \cos(t) \\ \alpha_6 \sin(t) + \alpha_3 \cos(t) - \cos(t) \end{pmatrix} \\
\alpha_4 &= \alpha_2 + \alpha_1 + 1 \\
-\alpha_1 &= \alpha_5 + \alpha_4 \\
\alpha_5 &= 2\alpha_3 - \alpha_2 \\
-\alpha_2 &= 2\alpha_6 - \alpha_5 \\
\alpha_6 &= \alpha_3 - 1 \\
-\alpha_3 &= \alpha_6
\end{aligned}$$